
Lights & Shadow



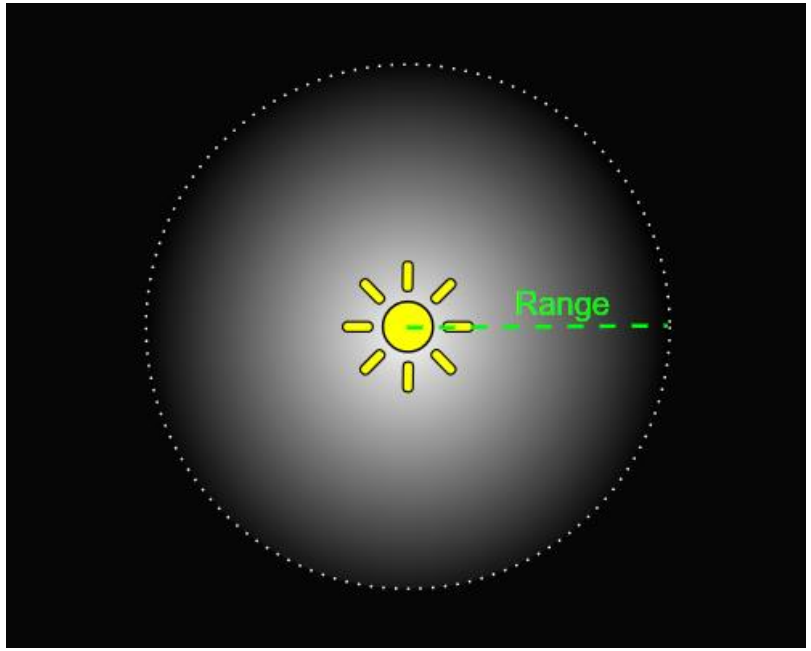
Range	10
Color	
Mode	Mixed
Intensity	1
Indirect Multiplier	1
Shadow Type	Soft Shadows
Baked Shadow Radius	0
Realtime Shadows	
Strength	1
Resolution	Use Quality Settings
Bias	0.05
Normal Bias	0.4
Near Plane	0.2
Cookie	None (Texture)
Draw Halo	
Flare	None (Flare)
Render Mode	Auto

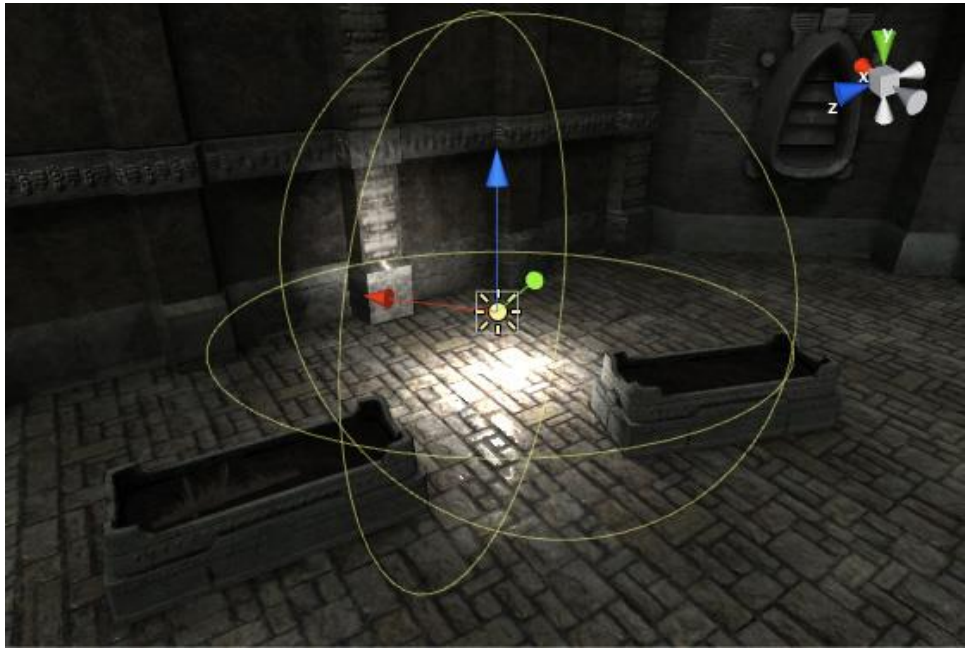
Lights and Shadows

Types of light

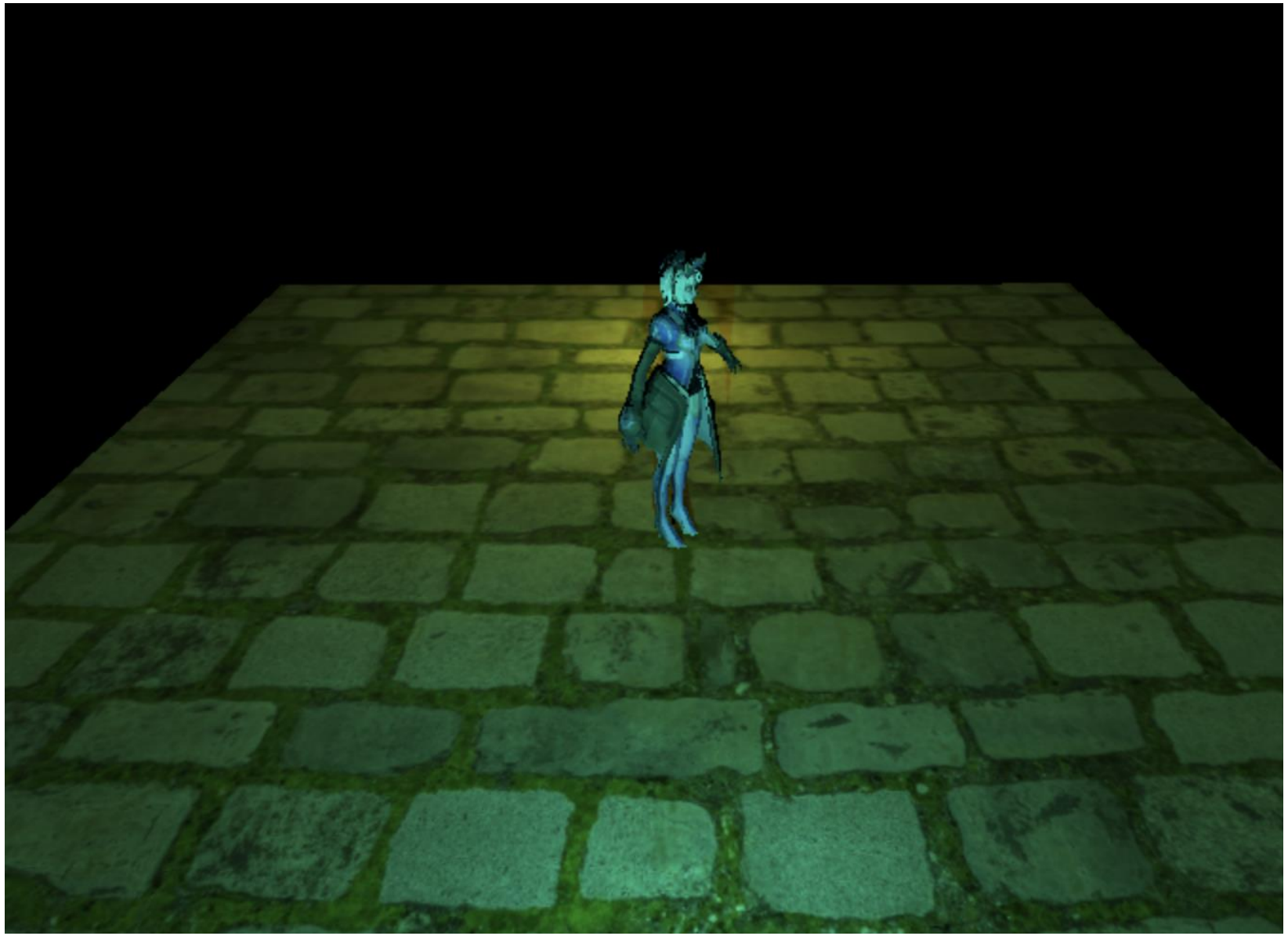
- **Point lights:** a point light is located at a point in space and sends light out in all directions equally.
- **Spot lights:** a spot light has a specified location and range over which the light falls off.
- **Directional lights:** a directional light behaving in many ways like the sun, directional lights can be thought of as distant light sources which exist infinitely far away.
- **Area lights:** an Area Light is defined by a rectangle in space. Light is emitted in all directions uniformly across their surface area, but only from one side of the rectangle.
- **Ambient Lights:** This light globally illuminates all objects in the scene equally.It cannot be used to cast shadows as it does not have a direction.

Point Lights

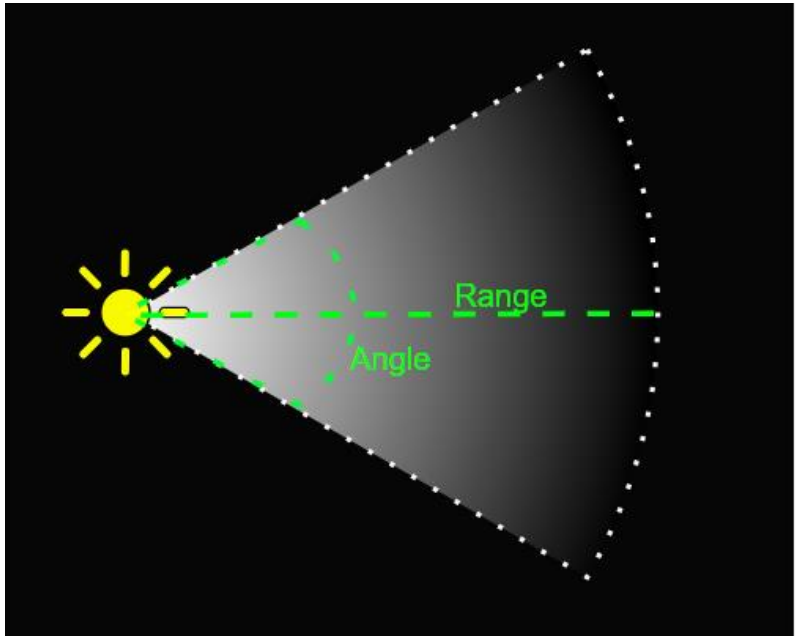


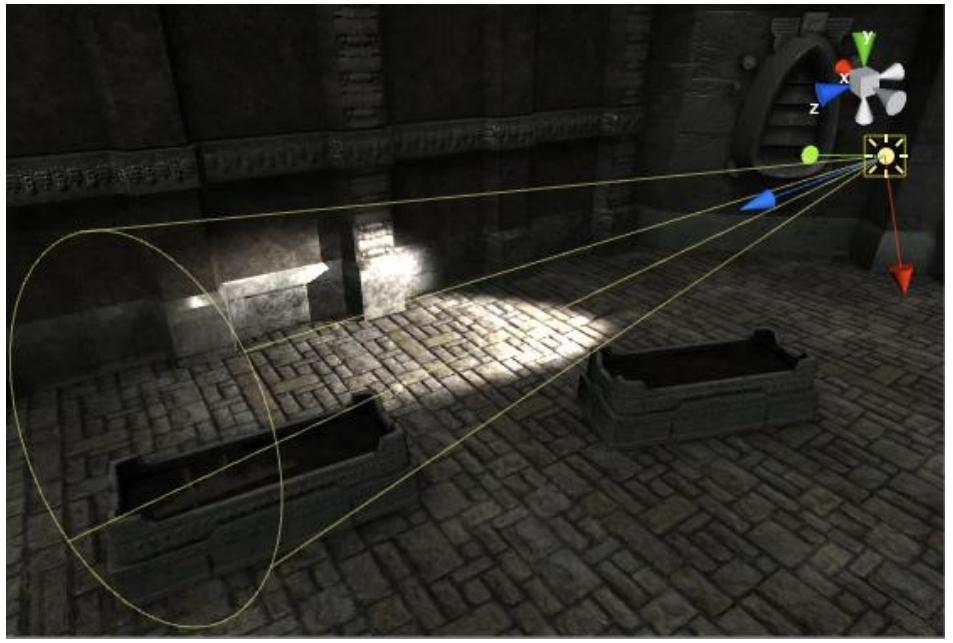


```
const pointLight = new THREE.PointLight(0xffffff, 2, 100);
pointLight.position.set(3, 0.5, 0);
pointLight.castShadow = true;
scene.add(pointLight);
```

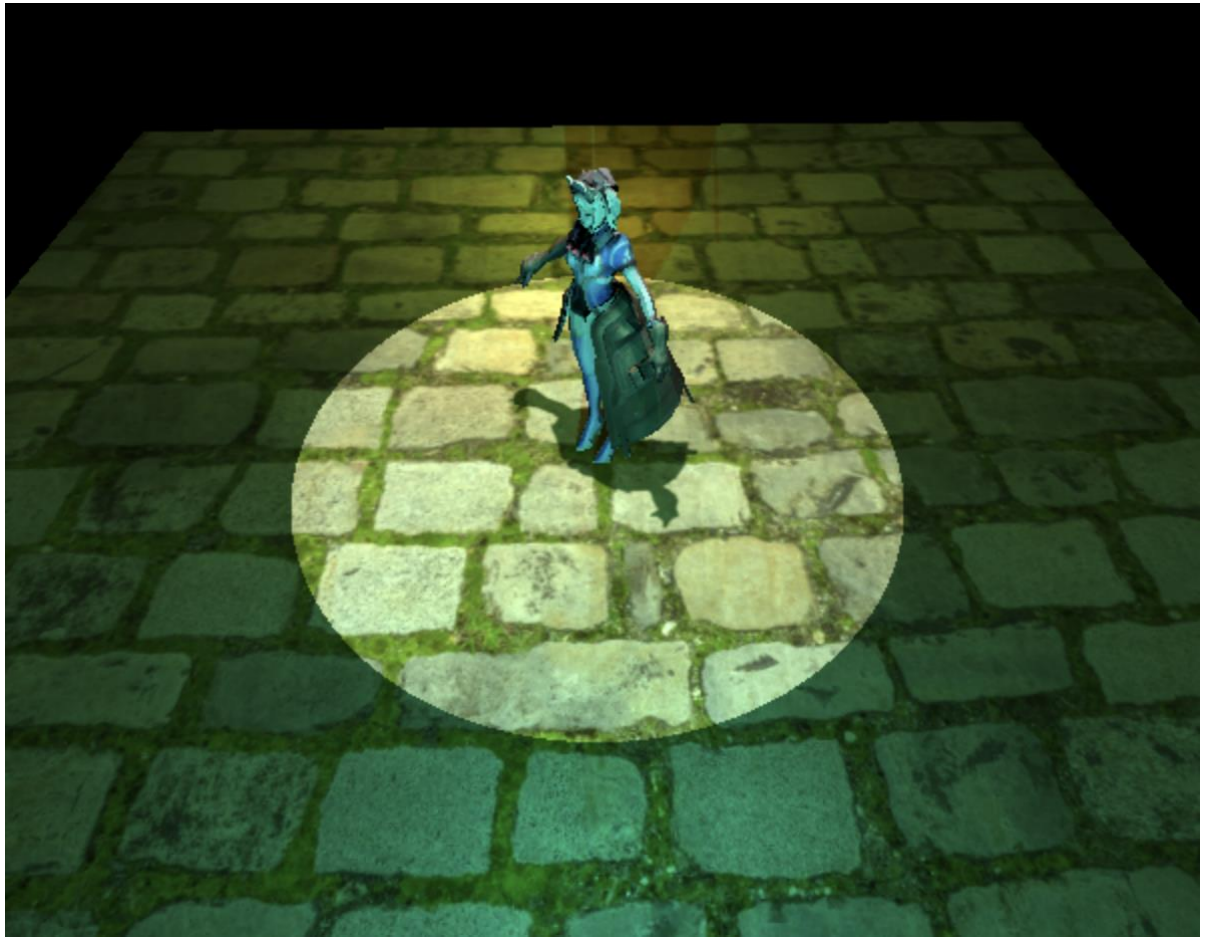


Spot Lights





```
const spotLight = new THREE.SpotLight(0xffffff, 1);
spotLight.position.set(-3, 1, 0);
spotLight.angle = Math.PI / 6;
spotLight.castShadow = true;
scene.add(spotLight);
```





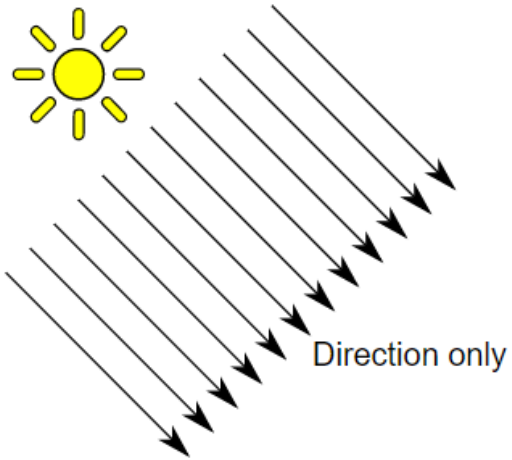
```
spotLight.position.set(0, 3, 0);
```

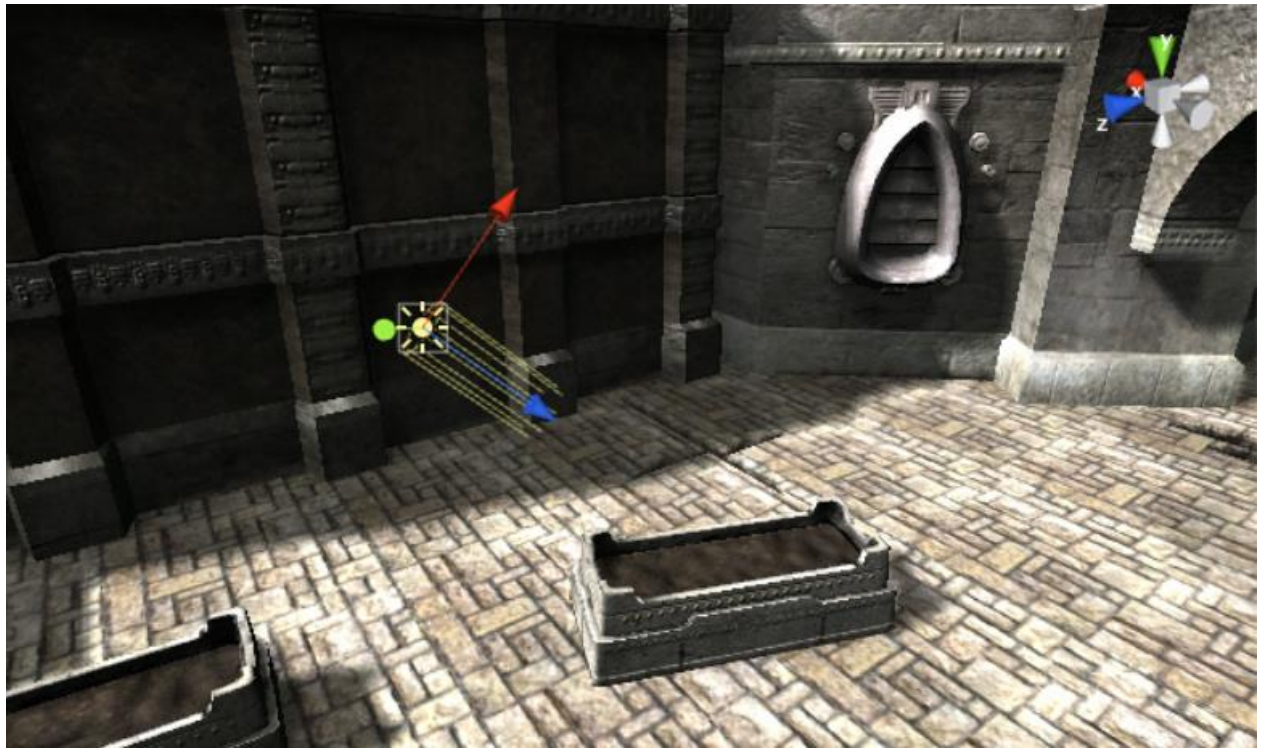
```
spotLight.position.set(3, 3, 3);
```



```
spotLight.position.set(-3, 3, -3);
spotLight.target.position.set(-3, 0, 0);
```

Direction Lights



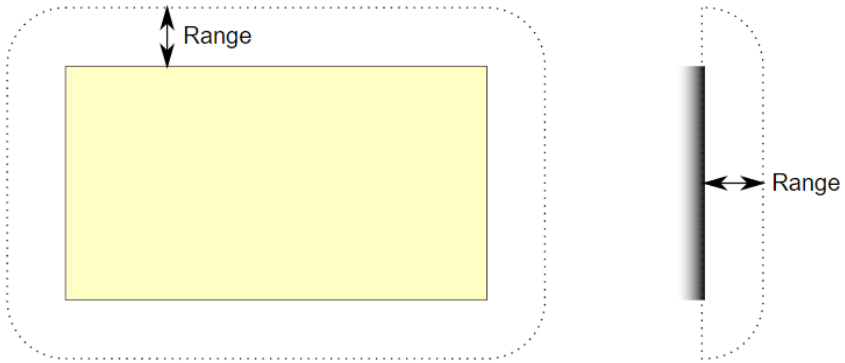


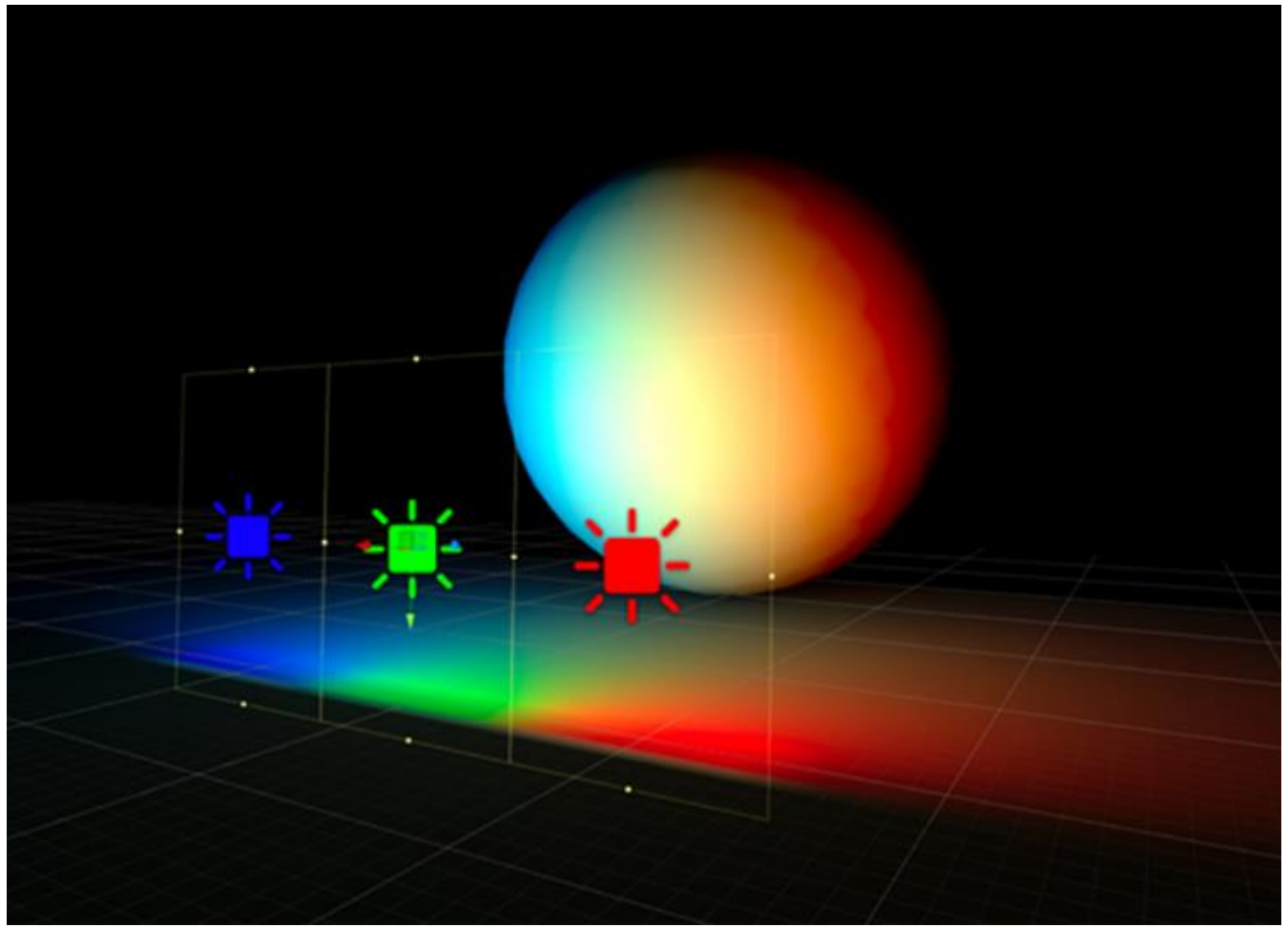
```
const directionalLight = new THREE.DirectionalLight(0xffeedd, 0.9);
directionalLight.position.set(10, 10, 10);
directionalLight.castShadow = true;
scene.add(directionalLight);
```



```
directionalLight.position.set(10, 10, 10);
```

Area Lights



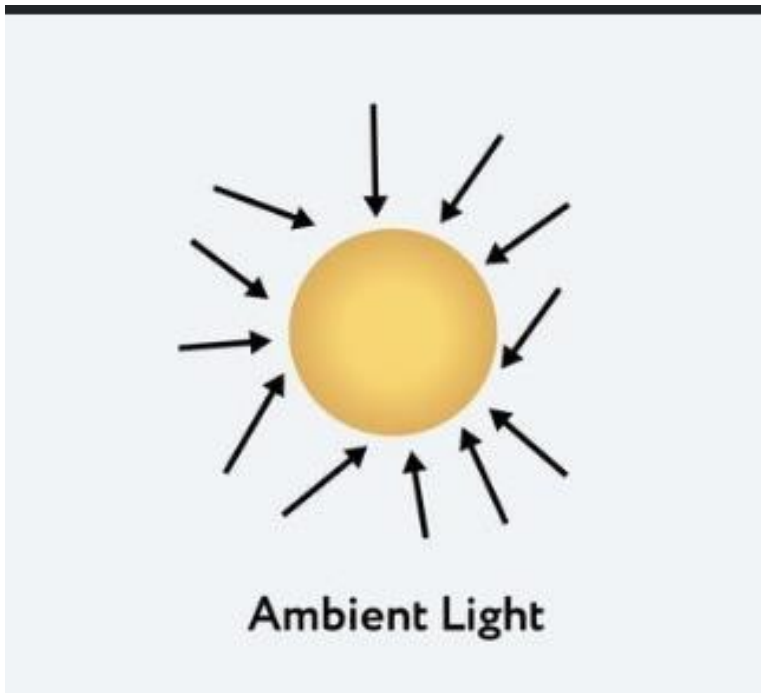


```
const areaLight = new THREE.RectAreaLight(0xffffff, 2, 2, 2);
areaLight.position.set(0, 1, 2);
areaLight.lookAt(0, 0, 0);
scene.add(areaLight);
```



```
const areaLight = new THREE.RectAreaLight(0xffffff, 1.5, 2, 2);
areaLight.position.set(3, 1, 3);
areaLight.lookAt(3, 1, 0);
```

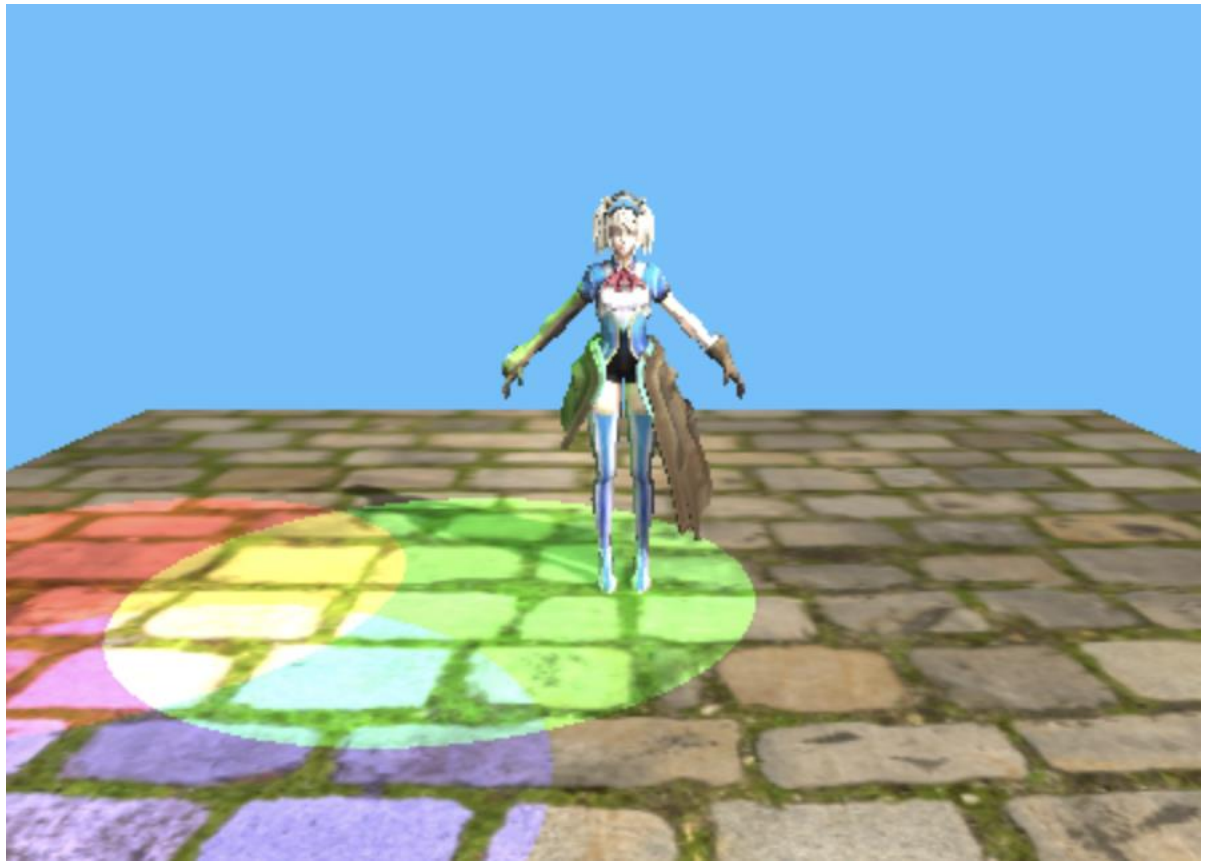
Ambient Lights





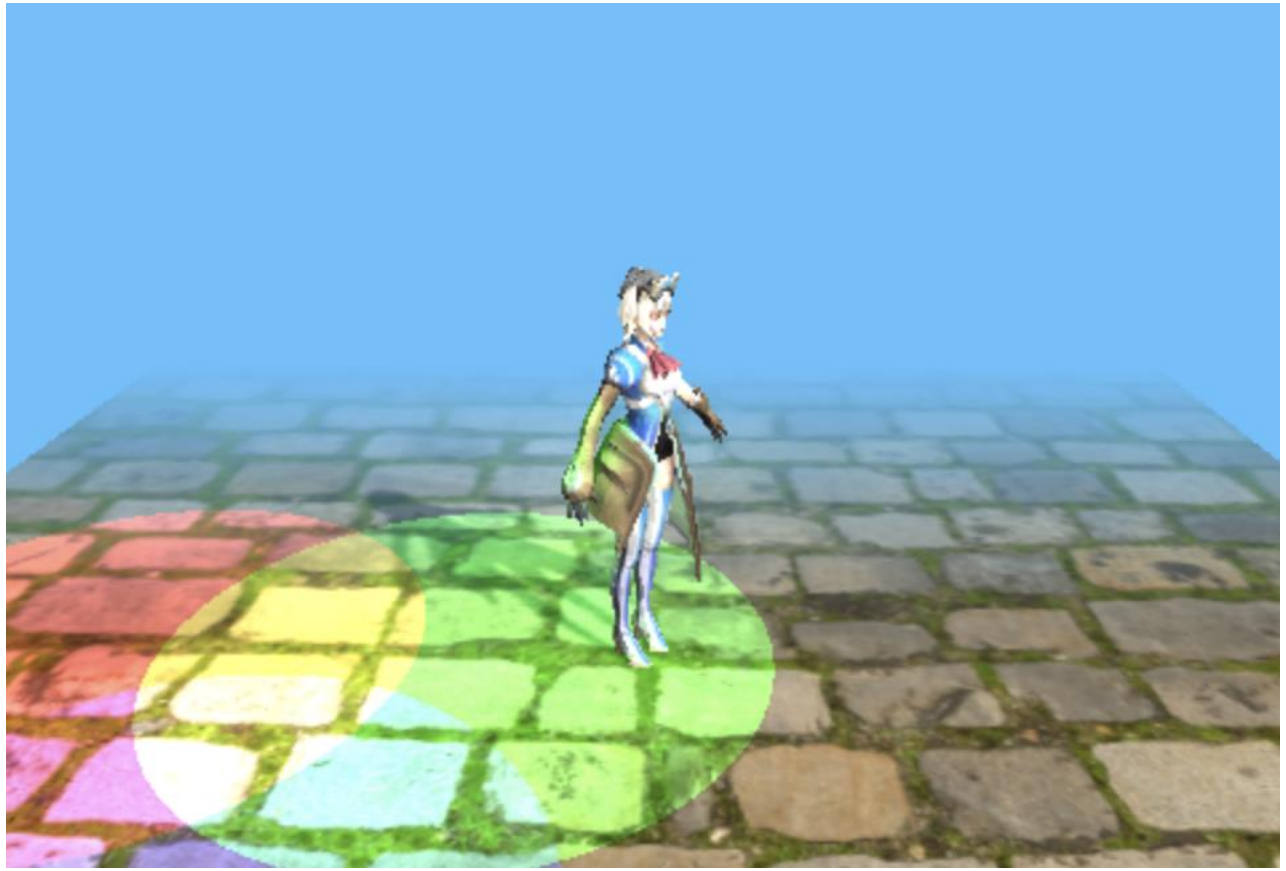
```
const abmientLight = new THREE.AmbientLight(0x404040, 1); // soft white light
scene.add(abmientLight);
```

Background



```
scene.background = new THREE.Color(0x5cc1ff);
```

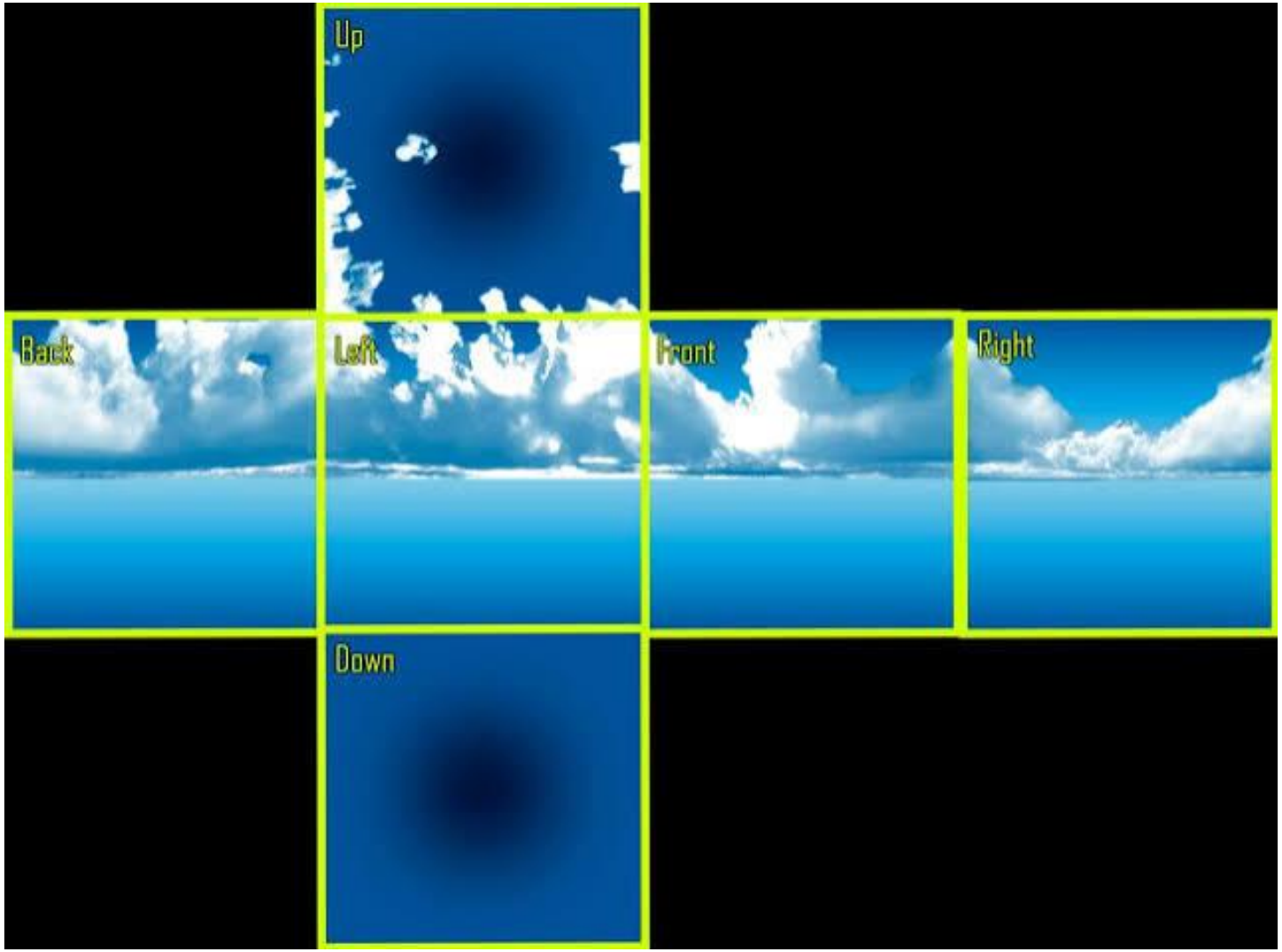
Fog



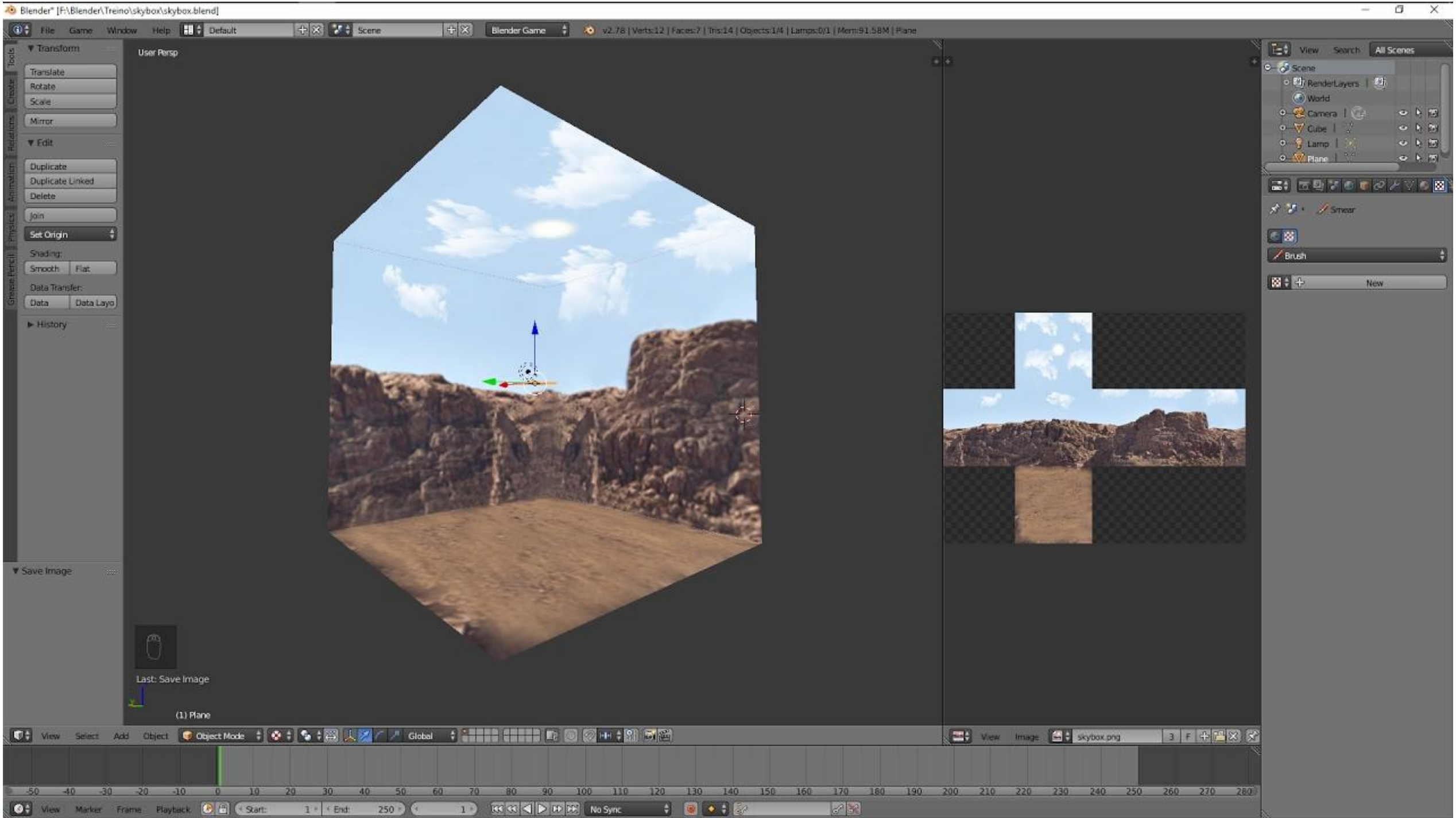
```
scene.fog = new THREE.Fog(0x5cc1ff, 5, 10);
```



Skybox




```
//skybox
const loader = new
THREE.CubeTextureLoader();
const skybox = loader
.setPath('https://raw.githubusercontent.com
/mrdoob/three.js/r128/examples/textures/c
ube/Bridge2/')
.load([
'posx.jpg', 'negx.jpg',
'posy.jpg', 'negy.jpg',
'posz.jpg', 'negz.jpg'
]);
skybox.encoding = THREE.sRGBEncoding;
scene.background = skybox;
scene.environment = skybox;
```


Blender* [F:\Blender\Treino\skybox\skybox.blend]
File Game Window Help Default Scene Blender Game
v2.78 | Verts:12 | Faces:7 | Tris:14 | Objects:1/4 | Lamps:0/1 | Mem:91.58M | Plane
Transform
Translate
Rotate
Scale
Mirror
Edit
Duplicate
Duplicate Linked
Delete
Join
Set Origin
Shading:
Smooth Flat
Data Transfer:
Data Data Layo
History
Save Image
User Persp
Last: Save Image
(1) Plane
View Select Add Object Object Mode Global
View Search All Scenes
Scene
RenderLayers
World
Camera
Cube
Lamp
Plane
Smear
Brush
New
View Image skybox.png 3 F
View Marker Frame Playback Start: 1 End: 250 1 No Sync

Top 30 Best Racing Games for PS1





https://www.youtube.com/watch?v=B9-cPI476mg

Paranoramic



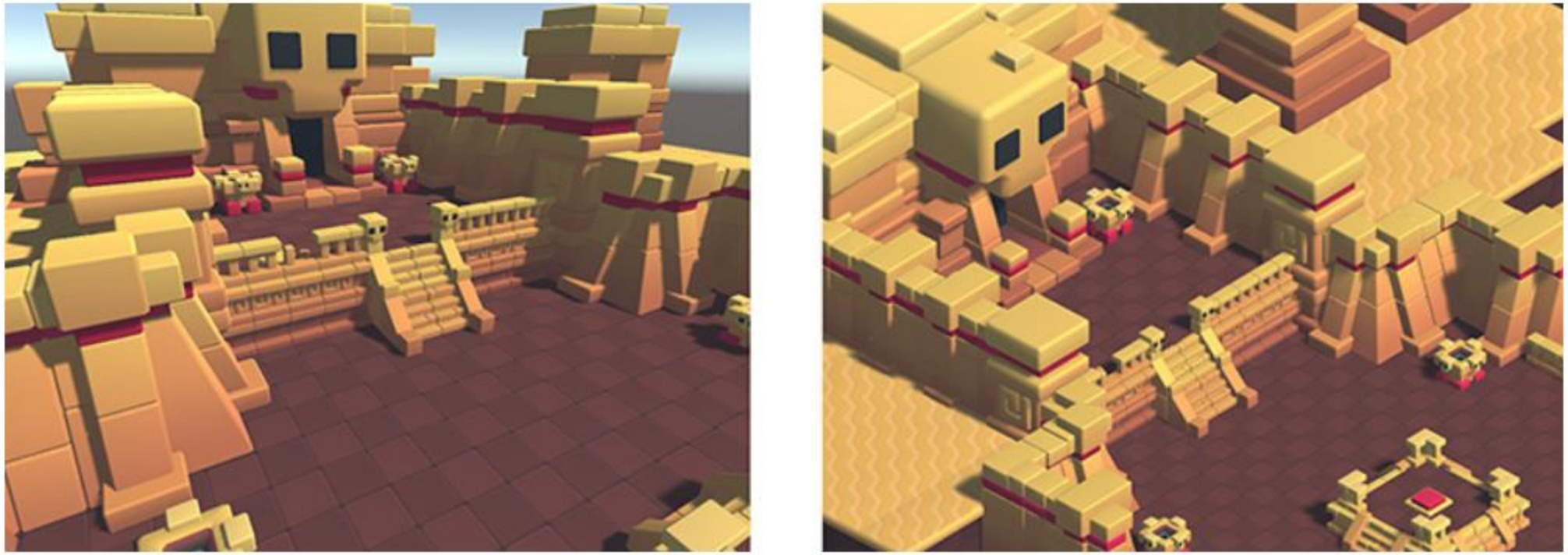
Bonus (GLTF,GLB) Loader

```
<script
src="https://cdn.jsdelivr.net/npm/three@0.128.0/examples/js/loaders/GLTFLoader.js">
</script>
```

Cameras

Cameras

- Perspective and orthographic (isometric) cameras



The same scene shown in perspective mode (left) and orthographic mode (right)

- Perspective and orthographic (isometric) cameras



